

Not so S(imple)NMP

Level 4: Step 1
NORMAL
Theme: Integrated Network Management
Due Date: 19/01/2023

Please take the following notes in consideration:

- In this assignment the questions marked with a (*) are practical and have to be shown to the teacher (no report is needed).
- The remaining questions are to be answered in a written report, to be delivered to the teacher.
- Whenever possible, the VMs necessary to resolve the practical questions should be created as *linked clones*.

Network Management

1. Describe and explain the five areas (FCAPS) of network management.
2. What data types are available in **SMIv1**?
3. Use the simpleweb.org **MIBs browser** to find out the **OID** for the `.ipAddressTable` object from the **IF-MIB**.

SNMP

4. (*) Create a **Debian Linux** VM and perform the following steps:
 - (a) Install the **Net-SNMP** agent service (`snmpd` package) and client tools (`snmp` package)
 - (b) Update the **MIBs** to their latest version using the `snmp-mibs-downloader` package (non-free)
 - (c) Create a **SNMPv3** user with **authentication** and **privacy**.
 - (d) Install **MRTG** (`mrtg` package) and configure it to show statistics from the **SNMP Agent** about:
 - Traffic Analysis
 - CPU load
 - Free Memory and Disk Usage

MRTG should have a sampling period of 5 minutes.
 - (e) Use the snmp client tools to get the **table** of running software on `localhost`
Hint: Take a look at the **OIDs** in the **HOST-RESOURCES-MIB**
5. (*) Create a VM with Windows and perform the following tasks:

- (a) Activate the Windows **SNMP Agent** (under **Optional Features**)
- (b) Modify the service to accept packets from any host and add the appropriate security communities for remote access
- (c) Install **CACTI** and configure it to show statistics from the **SNMP Agent** about:
 - Traffic Analysis
 - CPU load
 - Free Memory and Disk Usage
- (d) Use snmp client tools on the VM from item (4) to get the **table** of running software on this host (HOST-RESOURCES-MIB).

MIBs

6. Imagine you want to provide, via SNMP, information from **IPB's** Meteorological Weather Station:
 - (a) Use the syntax from **SMIv2** to write a **MIB** file to describe the information from an hypothetical **wind-gauge** and **wind-direction** (`[0; 360]`) sensor. Declare it under the `internet/private/enterprises` node (IETF RFC1155-SMI - **1.3.6.1.4.1**).
 - (b) Describe the data types and objects necessary and include the file in your report.

Deliverables

7. All deliverables submitted via [discord#assignments-submission](#)):
 1. PDF file with the report
 2. The file name should be `14s1n_aXXXXX_aYYYYY.pdf`. This code means:
 - Level 4
 - Step 1
 - Normal difficulty
 - Students that participated in the group
 - **Maximum:** 2 students